

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0565

Badoura 115 138 kV
April 18, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0565. The Interconnection Customer proposes a 170.4 MW wind facility interconnecting at the Badoura 115 138 kV substation in the Pennsylvania area. The request was received on January 03, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 273 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,515,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DEKALB 230-1 kV Autotransformer	RICHLAND - DEKALB 500.0 kV	117.6	1,082	6.6
BLUEBONNET - RUSSELL FORK 230.0 kV	DEKALB - JONES CHAPEL 500.0 kV	110.0	900	5.7
STATE ST - ROBISON PARK 230.0 kV	RICHLAND - DEKALB 500.0 kV	107.1	1,330	6.1
BLUEBONNET - JONES CHAPEL 230.0 kV Line	RUSSELL FORK - ROBISON PARK SUBSTATION 230.0 kV line	102.6	758	5.1
JONES CHAPEL 500 kV/230 kV Autotransformer	JONES CHAPEL - DEKALB 500.0 kV	100.2	1,617	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DEKALB 230-1 kV Autotransformer	Replace transformer with higher-rated unit	\$8,866,000	2% (\$152,000)
BLUEBONNET - RUSSELL FORK 230.0 kV	Rebuild line with higher-capacity conductor	\$62,444,000	2% (\$1,362,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STATE ST - ROBISON PARK 230.0 kV	Reconductor line	\$4,803,000	2% (\$105,000)
BLUEBONNET - JONES CHAPEL SUBSTATION 230.0 kV	Reconductor line	\$12,276,000	2% (\$266,000)
JONES CHAPEL 500 kV/230 kV Autotransformer	Replace transformer with higher-rated unit	\$28,784,000	2% (\$630,000)
	Total allocated to project		\$2,515,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.